

Essential Criteria

Qualifications, experience and knowledge

Possession of a PhD or equivalent qualification in a relevant subject area I hold a PhD in Electrical Engineering from the [University of Southern California](#), completed within its [Signal and Image Processing Institute](#).

An innovative personal research agenda in NeuroAI, computational neuroscience, or a A/I closely related field. Please refer to my Research Vision in rapelaVisionStmnt.pdf.

Evidence of the ability to deliver and publish internationally excellent research. Publications should demonstrate both methodological rigour (e.g. development or application of quantitative/mathematical frameworks) and engagement with genuine biological or neuroscientific questions, not purely technical contributions to AI/ML alone. I have followed a non-traditional career path in computational neuroscience. Following a PhD and two postdoctoral fellowships at the interface of statistical machine learning and neuroscience—resulting in key publications in leading journals such as *Neural Computation*, *Network: Computation in Neural Systems*, and the *Journal of Vision*—I chose to embed myself within a world-class computational/experimental environment. I joined the Gatsby Computational Neuroscience Unit and the SWC as a Research Engineer Fellow to further develop my skills in statistical machine learning, software engineering and team science.

I have grown enormously at the Gatsby and SWC. At the Unit, I have substantially increased my expertise in signal processing and machine learning, specially through my weekly meetings with Prof. Maneesh Sahani for more than seven years. Alongside developing professional software engineering capabilities, I have been an active contributor to the institute-wide team science Aeon project. My leadership and research contributions include securing a £800,000 BBSRC grant as well as co-authoring a joint international application between the Gatsby Unit, the SWC, and the Allen Institute for Neural Dynamics for a BBSRC/NSF-Bio award. Over the past three years, I have successfully managed a small team of research software engineers, designed and taught the core Statistical Neuroscience course for SWC PhD candidates, and mentored two undergraduate fellows through year-long Simons Foundation fellowships.

While my primary focus during this period has been on software infrastructure and machine learning training, my next career phase will centre on publishing research articles, teaching and mentoring students, and establishing active collaborations with experimental groups at NPP and CDB. I am well prepared for this transition; over the last seven years, I have developed several research projects that will yield high-quality publications in the near future. Below, I outline a few of these upcoming papers:

- **Traveling waves during rhythmic speech production:** Expanding on single-subject findings reported in my earlier work (Rapela 2016, 2017, 2018), I am leveraging a newly released dataset across six subjects to validate these spatio-temporal dynamics in a larger population.
- **Statistical and software foundations of svGPFA:** I am preparing a pair of manuscripts documenting the unique numerical, statistical, and software engineering features of our Python distribution of svGPFA. These will be accompanied by a scientific paper demonstrating svGPFA findings on interesting neuroscience problems, such as detecting replay in the striatum.
- **Algorithmic extensions to svGPFA:** To broaden the scope of the framework, I am extending svGPFA to incorporate external inputs and support online operation.
- **Bonsai.ML and high-throughput behavioural analytics:** I will introduce the *Bonsai.ML* package for intelligent, closed-loop experimental control. Additionally, in collaboration with a talented undergraduate mentee, we are preparing a report on the statistical analysis of long-duration, continual mouse behavioural data within the Aeon platform.
- **State-space models for kinematics inference:** To address the failure of deep pose-estimation networks to account for temporal continuity, we have developed [state-space models that explicitly infer kinematics](#). I will describe these methods and demonstrate their advantages in robustly capturing single- and multi-animal behaviour.

Evidence of potential to secure competitive research funding. During my postdoctoral position at UC San Diego I was awarded a \$30,000 seed grant from the *UCSD Center for Brain Activity Mapping* and at the Gatsby Unit an £800,000 grant from BBSRC.

Evidence of effective supervision and mentoring of undergraduate, postgraduate and/or doctoral research students including teaching in a computational, data science, or quantitative context (e.g. programming, machine learning, statistics, mathematical modelling) — ideally in neuroscience or bioscience settings; Python experience strongly preferred. As part of the [Simons Foundation’s SURFiN program](#) I mentored an undergraduate student from QMUL and another from UCL’s NPP. I have been teaching assistant, discussion leader, course coordinator and instructor in courses in Python programming, artificial intelligence, numerical linear algebra, probability, statistical neuroscience, delivered to undergraduate and graduate engineering and neuroscience students. Please refer to my CV for details.

Evidence of productive cross-disciplinary collaborations, specifically spanning computational or AI methods and neuroscience or biological systems, for example joint A/I publications, grants, or collaborative projects with researchers from complementary disciplines. I owe much of my research trajectory in computational neuroscience to close partnerships with experimental collaborators:

- During my PhD, I collaborated with Dr. John Touryan and Dr. Gidon Felsen (UC Berkeley, Yang Dan lab) to characterise responses to natural images in the cat primary visual cortex, utilising multivariate nonlinear regression models that I developed ([Rapela et al., 2006, 2010](#)).
- During my first postdoc at UC San Diego, a collaboration with Prof. Edward Chang (UCSF) helped me secure my first seed grant. His lab generously shared human ECoG recordings from epileptic subjects, which I used to fit and evaluate new ensemble dynamical models ([Rapela et al., 2015](#)). This rich dataset ultimately enabled me to discover travelling waves in human language regions during speech production ([Rapela, 2016, 2017, 2018](#)).
- Also at UC San Diego, I collaborated with EEG experimentalists Prof. Jeanne Townsend and Dr. Marissa Westerfield to study the relationship between single-trial phase coherence and attentional shifts in humans ([Rapela et al., 2018](#)). The machine learning innovation in this work was the use of variational inference to estimate large-dimensional linear regression models from limited data.
- At the SWC and Gatsby Unit, I also contributed statistical expertise to the analysis of high-throughput behavioural data within the Aeon platform ([Campagner et al., 2025](#)). Through this initiative, I partnered with Dr. Goncalo Lopes (Director of NeuroGEARS), with whom I co-secured and developed the BBSRC-funded project [Machine Intelligence for Neuroscience Experimental Control](#).

Experience of engagement beyond academia, including contributions to policy, practice, public engagement, or knowledge exchange. I have facilitated knowledge exchange through the translation of complex machine learning into accessible, open-source scientific tools. I have focused on bridging the gap between machine learning and its practical, real-world application by experimental scientists.

- **Knowledge Exchange and Community Building:** As a co-developer of the *Bonsai.ML* ecosystem and the statistical infrastructure for the *Aeon* project, I have facilitated knowledge exchange across scientific domains. An example of this engagement was my involvement in the **2024 Bonsai Developers Conference**, where we brought together an international community of developers and experimentalists to share insights, standardise workflows, and lower the barrier to integrating machine learning into laboratory environments.
- **Transforming Practice via Open Software Distribution:** I have developed and openly distributed statistical packages to the wider scientific community. By translating complex mathematical concepts into easy-to-use software, I have enhanced computational workflows across laboratories. My publicly distributed contributions include:
 - [joacorapela/svGPFA](#): Python implementation for Sparse Variational Gaussian Process Factor Analysis.
 - [joacorapela/ssm](#): State-space models for behavioural and neural data processing, offline and in real time.
 - [joacorapela/hiddenMarkovModels](#): R implementations for HMMs with multivariate Gaussian, Poisson, Poisson GLM, and autoregressive observations.
 - [joacorapela/bayesianLinearRegression](#) & [joacorapela/variationalLinearRegression](#): Python code for exact and variational Bayesian inference.
- **Contributions to Educational Practice:** I translate research into educational practice. I designed and taught *Statistical Neuroscience* course for PhD students at the SWC and mentored two Simmons Foundation fellows. This work represents a commitment to practice, ensuring that the next generation of biomedical and NeuroAI researchers are equipped with strong quantitative skills.

Skills and abilities

Ability to develop and implement team-based solutions to complex challenges, delivering positive outcomes. Team science has been central to my computational neuroscience career. I believe that the most complex challenges in computational neuroscience cannot be solved in isolation; they require the integration of diverse perspectives, technical skill sets, and experimental methodologies. I have approached scientific obstacles by implementing team-based solutions with my collaborators.

- **Bridging Theory and Experimentation:** At the Gatsby Unit and the SWC, I have helped connect statistical methods with neuroscience experimentation. Working in partnership with researchers, engineers and data scientists, I have co-developed software infrastructure like *Bonsai.ML* and the *Aeon* project. These platforms are excellent and successful team-science results.
- **Delivering Major Collaborative Outcomes:** This team-first approach has delivered verifiable, positive academic results. I was a primary driver in securing and executing an £800,000 BBSRC grant. Both the grant application and its development are excellent examples of team-science in action.

As a faculty member, I will build this culture of team science directly into group, and its interaction with NPP, CDB and other collaborators.

Advanced computational skills relevant to neuroscience research, including proficiency in at least one scientific programming language (e.g. Python, MATLAB, R, Julia) and familiarity with machine learning frameworks or data analysis tools used in neuroscience or bioscience. I possess advanced computational skills in signal processing, optimisation, machine learning, statistics, computer science and software development. I built these skills in my undergraduate and graduate studies, postdoctoral training, my current research engineer fellow role at the Gatsby Unit, and in my software development roles in industry. I am highly proficient in both Python and R for data analysis, statistical modelling, and scientific visualisation. I have engineered and optimised the core architecture of the *svGPFA* package utilising both PyTorch and JAX backbones.

UCL Ways of Working

Ability to build collaborative research and industry partnerships. My research career in computational neuroscience provides evidence of my ability to establish and sustain research partnerships. My work naturally exists at the intersection of diverse disciplines, requiring continuous synthesis between theoretical modelling and experimental application.

Beyond traditional academic networks, I have also built collaboration with industry partners. An example is my successful collaboration with *NeuroGEARS Ltd.* Together, we co-developed the strategic vision, successfully bid for funding, and executed the *Bonsai.ML* grant.

As a faculty member at UCL, I intend to leverage these established industry and academic networks to drive innovative, well-funded cross-sector collaborations.

Ability to develop and implement team-based solutions to complex challenges, delivering positive outcomes. Answered above.

Active commitment to equality, diversity, and inclusion, for example through inclusive teaching practices, mentoring underrepresented students, participating in EDI initiatives, or fostering an inclusive lab/group culture. Having been a member of an underrepresented minority in the US and the UK, I am highly sensitive to EDI issues. This was one of my motivations for participating in the SURFiN mentoring program of the Simons Foundation, where I mentored a first student from Pakistan and a second one from China.

I am deeply dedicated to fostering an explicitly inclusive research and classroom culture. In a faculty role at UCL, I will leverage my experiences to implement equitable teaching practices in quantitative courses and actively recruit, support, and advocate for diverse talent within the joint environment of NPP and CDB.

Desirable Criteria

Postgraduate Certificate in Teaching & Learning in Higher Education, or equivalent qualification. While I do not currently hold a formal Postgraduate Certificate in Higher Education, I am highly motivated to obtain this qualification and would actively pursue it upon appointment. I have extensive practical experience in educational design and delivery, having created and taught the foundational *Statistical Neuroscience* course for PhD students at the SWC

and mentored international fellows. I view the opportunity to complete a formal teaching certificate at UCL as a valuable mechanism to ground my existing pedagogical experience in structured higher-education theory and to ensure my classroom methods remain inclusive and highly effective.

Experience of contributions to peer review bodies/committees, professional organisations, learned societies, government committees or Research Councils etc. I am an ad-hoc reviewer for Vision Research, Frontiers in Neuroscience, and Journal of Perceptual Imaging. In addition, I have reviewed grant proposal for the Software Sustainability Institute.

Experience of an active role in widening participation initiatives that span research communities, e.g. sharing ideas and experience through organising seminars and workshops. I believe that advancing scientific discovery requires building open, transparent, and equitable research environments. To actively contribute to this mission, I served as a member of the SWC [Research Culture Working Group \(RCWG\)](#).

In this role, I collaborated with a cross-disciplinary team of researchers and staff to evaluate and improve the research culture of our institute. By designing initiatives that encourage the open sharing of data, software workflows, and technical expertise across distinct scientific communities, this work directly advanced widening participation, knowledge equity, and intellectual accessibility within the broader SWC and Gatsby Unit ecosystem.

References

- Campagner, D., Bhagat, J., Lopes, G., Calcaterra, L., Pouget, A., Almeida, A., Nguyen, T., Lo, C., Ryan, T., Cruz, B., et al. (2025). Aeon: an open-source platform to study the neural basis of ethological behaviours over naturalistic timescales. *bioRxiv*, pages 2025–07.
- Rapela, J. (2016). Entrainment of traveling waves to rhythmic motor acts.
- Rapela, J. (2017). Rhythmic production of consonant-vowel syllables synchronizes traveling waves in speech-processing brain regions.
- Rapela, J. (2018). Traveling waves appear and disappear in unison with produced speech.
- Rapela, J., Felsen, G., Touryan, J., Mendel, J. M., and Grzywacz, N. M. (2010). ePPR: a new startegy for the characterization of sensory cells from input/output data. *Network: Computation in Neural Systems*, 21(1-2):35–90.
- Rapela, J., Mendel, J., and Grzywacz, N. (2006). Estimating nonlinear receptive fields from natural images. *Journal of Vision*, 6(4):441–474.
- Rapela, J., Rowat, P. F., Kostuk, M., Mullen, T., Chang, E. F., and Bouchard, K. E. (2015). Modeling neural activity at the ensemble level.
- Rapela, J., Westerfield, M., and Townsend, J. (2018). A new single-trial foreperiod effect on inter-trial phase coherence. Part I: existence and relevance. *Neural Computation*, 30(9):2348–2383.